

Beyond Flat Chunking: The Paradigm Shift Toward Unified Visual-Structural Representations in Multimodal RAG

The integration of Retrieval-Augmented Generation (RAG) with Multimodal Large Language Models (MLLMs) marks a decisive shift from brittle, text-centric pipelines toward holistic document intelligence. Traditional approaches relying on Optical Character Recognition (OCR) and flat chunking frequently discard critical spatial fidelity and layout-dependent semantics, leading to fragmented evidence retrieval [1]. In response, emerging frameworks advocate for unified visual representations and graph-based indexing to preserve structural integrity. Vision-first methods like ColPali and VisRAG demonstrate that embedding documents as images bypasses parsing errors, though debates persist regarding their generalization compared to optimized textual baselines [233] [127]. Concurrently, layout-aware systems such as LAD-RAG and HiKEY reconstruct logical dependencies through symbolic graphs, enabling dynamic, query-adaptive retrieval that overcomes the limitations of isolated segments [32]. Despite these advancements, challenges remain in modality bias, where models under-utilize visual evidence, and in the scalability of fine-grained retrieval [3]. Future progress hinges on agentic workflows and rigorous evaluation metrics that ensure faithfulness and precise visual attribution across diverse, multilingual domains [80].

Introduction to Multimodal RAG in Document and Chart Understanding

The integration of Retrieval-Augmented Generation (RAG) with Multimodal Large Language Models (MLLMs) signifies a fundamental paradigm shift from traditional text-centric processing toward holistic reasoning across text, charts, tables, and complex layout structures. Historically, document intelligence relied heavily on Optical Character Recognition (OCR) pipelines to convert visually rich documents (VRDs) into flat text before indexing, yet this approach frequently suffers from brittle extraction processes that discard spatial fidelity, visual context, and the intricate relationships between textual and non-textual elements [1]. Conversely, native MLLMs often struggle with the computational constraints of long-context modeling when processing entire documents without retrieval mechanisms, leading to information interference and inefficiency [62]. Consequently, Multimodal RAG has emerged as a critical solution to synthesize information from VRDs through grounded retrieval, addressing the limitations of both flat semantic retrieval and isolated modality processing [7]. Current research underscores the necessity of evolving from simple chunking to sophisticated approaches leveraging graph structures, agentic frameworks, and unified visual representations to capture dense textual content and fine-grained semantic dependencies [4].

A primary challenge in this domain is the loss of structural information inherent in standard OCR-based workflows, where methods converting images to text prior to indexing often fail to preserve the layout-dependent semantics crucial for interpreting tables and figures [83]. In contrast, vision-based approaches that embed document pages directly as images have demonstrated significant advantages by bypassing OCR entirely. Systems like VisRAG utilize Vision-Language Models (VLMs) to generate embeddings that retain original visual cues, thereby eliminating information loss during parsing and achieving substantial performance gains over text-based baselines [113]. Similarly, ColPali employs a late interaction mechanism to produce high-quality multi-vector embeddings from document images, largely outperforming modern document retrieval pipelines while being drastically simpler and end-to-end trainable [233]. Empirical comparisons suggest that direct multimodal embedding retrieval often outperforms LLM-summary-based approaches, particularly in domains where numerical precision and spatial relationships are paramount [53]. However, contradictions exist within the literature; while vision-based methods excel on seen distributions, some studies indicate that OCR-based systems may still offer better generalization to unseen document qualities, highlighting a trade-off between structural fidelity and robustness [127]. Furthermore, recent analyses suggest that improvements in multilingual benchmarks may be driven more by better document representation than by the retrieval mechanism itself, calling for decomposed evaluation strategies [39].

To address the fragmentation of evidence across multi-page documents, recent frameworks have introduced graph-based and layout-aware mechanisms to overcome the limitations of isolated chunking. Conventional RAG systems often encode content in disconnected segments, severing cross-page dependencies and layout structures, whereas LAD-RAG constructs a symbolic document graph alongside neural embeddings to enable dynamic, query-adaptive retrieval [32]. Similarly, approaches like MMGraphRAG and DSRAG integrate multimodal knowledge graphs to capture logical chains and entity relationships across modalities, facilitating deeper reasoning than flat vector retrieval allows [155] [56]. These graph-enhanced methods allow for the modeling of hierarchical organization and inter-element relationships, such as figure-text correspondence, which are essential for human-like comprehension [2].

Granularity of retrieval is another pivotal area of innovation, as coarse-grained, page-level retrieval often introduces irrelevant context that inflates token usage and confuses generators. Frameworks like LFRAG and HiKEY advocate for fine-grained, block-level or hierarchical retrieval, segmenting documents into semantically coherent units that align with layout structures to enable precise query-content alignment [8] [31]. This shift significantly reduces token consumption during generation while improving retrieval accuracy. Furthermore, agentic frameworks are redefining the retrieval process by introducing iterative reasoning and dynamic planning. Systems such as Multimodal DeepResearcher and ViDoRAG employ multi-agent workflows to decompose complex queries, iteratively refine search strategies, and synthesize interleaved text-chart reports, demonstrating superior performance in deep research tasks [157] [80].

Despite these advancements, contradictions and gaps remain regarding evaluation and modality fusion, as some benchmarks suggest visual retrievers outperform textual ones while others indicate the gap is narrowing or that hybrid approaches yield the best results depending on the domain [48] [95]. Additionally, the handling of charts and specialized visual elements remains a bottleneck, with many MLLMs exhibiting a text-over-visual bias even when

ground-truth retrieval is provided [3]. The field is increasingly moving toward unified frameworks like UniversalRAG and RAG-Anything, which aim to handle heterogeneous modalities and granularities within a single architecture, yet challenges in efficiency, fine-grained representation, and robust cross-modal reasoning persist [247] [46]. Future progress hinges on developing adaptive retrieval units, reducing model sizes for deployment, and establishing comprehensive benchmarks that accurately reflect the complexity of real-world multimodal document understanding [7].

The Limitations of Text-Centric and Flat Retrieval Paradigms

Current document question answering systems predominantly rely on flattening documents into disconnected text chunks, a practice that fundamentally disrupts the logical and spatial dependencies inherent in visually rich documents. This approach neglects intrinsic hierarchical structures, leading to significant performance degradation when reasoning over complex visual formats such as charts, multi-page tables, and interleaved figures. Conventional retrieval-augmented generation methods encode content in isolated chunks during ingestion, thereby losing structural and cross-page dependencies essential for accurate interpretation [32]. Representing documents as a set of disconnected text chunks disrupts logical and spatial dependencies, such as section organization, figure-text correspondence, and cross-reference relations, which humans naturally exploit for comprehension [2]. The standard pipeline, which typically involves Optical Character Recognition followed by text-based embedding, often discards critical layout cues and visual semantics, resulting in incomplete evidence retrieval and degraded answer quality for multi-hop reasoning tasks [294].

The limitation of flat retrieval is further exacerbated by the text-over-visual modality bias observed in many standard implementations. Models trained primarily on textual corpora tend to ignore retrieved visual evidence in favor of textual cues, even when the visual component contains the definitive answer. Even state-of-the-art multimodal large language models demonstrate consistent text-over-visual modality bias during chart-based reasoning, often failing to utilize the visual data despite its availability [3]. Strong modality biases and the under-utilization of retrieved visual evidence have been revealed across various open-source and proprietary models, indicating a systemic failure to integrate multimodal signals effectively [11]. When documents are reduced to plain text, the spatial logic and layout cues necessary for reasoning are lost, a phenomenon described as visual semantic loss, where converting visually rich documents into text discards the spatial logic essential for complex reasoning [148]. Furthermore, while some approaches attempt to treat documents as images, they often fail to leverage underlying structured sources effectively, yet purely text-based representations remain insufficient for figure-based queries [172].

The flattening paradigm fails to capture the hierarchical and relational structures that are critical for document-level reasoning. Existing methods struggle because extracted elements are often fragmented and stripped of their original semantic context, lacking effective representations to capture hierarchical structures within documents [294]. The inability to model these relationships leads to evidence fragmentation, where scattered information across multiple pages cannot be organically connected within a limited token budget [31]. This issue is particularly acute in scientific and industrial documents where critical evidence is distributed across modalities and pages, and comprehending hierarchical semantic relations among multiple pages is essential to locate multimodal components [69]. Flat chunking yields semantic fragments that are hard to interpret in isolation, while iterative retrieval often fails by looping on partial evidence or drifting into irrelevant sections due to the lack of a persistent global search state [66]. Traditional text-based chunking methods struggle with complex document structures, multi-page tables, and contextual dependencies across page boundaries, necessitating a shift toward vision-guided approaches that maintain semantic coherence [20].

The reliance on text-centric paradigms also introduces brittleness in the face of diverse document layouts and formats. Traditional methods that deal with flat text chunks stand in contrast to layout-aware graph modeling, which retains the structure of input documents by connecting text chunks, tables, and figures [55]. Without such structural retention, systems cannot handle complex questions requiring information from multiple modalities. Combining robust visual retrieval with linguistic reasoning through a consistency-constrained modality fusion mechanism significantly outperforms unimodal baselines, underscoring the inadequacy of treating text and vision as separate or flattened entities [9]. The prevailing text-centric and flat retrieval paradigms create a fundamental bottleneck in document understanding, necessitating a paradigm shift toward structure-aware, multimodal, and hierarchical retrieval frameworks that can preserve the rich interplay between text, layout, and visual elements.

The Emergence of Unified Visual and Structural Representations

The traditional paradigm of document understanding, which relies heavily on parsing documents into linear text streams, is increasingly recognized as insufficient for capturing the rich, multimodal nature of modern information sources. A primary limitation of classical approaches is the inherent information loss that occurs when complex layouts, visual cues, and structural relationships are discarded during the conversion to plain text. To address this fragmentation, recent research advocates for treating documents as unified image formats or structured graphs, thereby preserving the integrity of the original content. This shift is driven by the realization that a significant portion of enterprise data resides in unstructured, heterogeneous files where visual context is paramount for accurate interpretation [196]. By compressing visual information into dense token representations and aligning them with textual content, new frameworks prevent the degradation of semantic fidelity associated with traditional optical character recognition pipelines [6].

A central theme in this emerging landscape is the move toward parsing-free methodologies that leverage Vision-Language Models to process document screenshots directly. Approaches such as Document Screenshot Embedding demonstrate that encoding entire document pages as images allows retrieval systems to bypass brittle text extraction steps, effectively capturing layout nuances, font styles, and graphical elements that text-only models miss [141]. Similarly, VisRAG establishes a pipeline where documents are embedded as images to maximize the retention of original data information, eliminating the errors introduced by intermediate parsing stages [113]. However, while treating documents as images preserves visual fidelity, it can sometimes obscure fine-grained textual semantics or create storage bottlenecks due to the high dimensionality of patch-based embeddings. To mitigate this, hybrid strategies like ColParse generate a small set of layout-informed sub-image embeddings fused with global page vectors, achieving a balance between structural awareness and computational efficiency [271]. Furthermore, Unveil proposes a knowledge distillation framework that transfers semantic understanding from visual-textual models to purely visual ones, enabling efficient retrieval without sacrificing the ability to comprehend text-rich scenarios [291].

Parallel to the adoption of unified image representations is the construction of explicit structural maps and knowledge graphs that encode inter-element relationships. This graph-based approach addresses the limitation of flat chunking, which often severs the logical connections between figures, tables, and their corresponding textual descriptions. Docs2KG introduces a framework that dynamically generates unified knowledge graphs from heterogeneous documents, facilitating efficient querying and exploration of data lakes that would otherwise be inaccessible to keyword-based search [196]. In the domain of financially dense table-text documents, Doc2SoarGraph employs semantic-oriented hierarchical graphs to capture correlations among elements, significantly outperforming token-level methods in discrete reasoning tasks [151]. The utility of graph structures extends to chart understanding as well, where ChartKG models visual elements and their semantic relations within chart images to support downstream tasks [98]. Additionally, MMGraphRAG refines visual content through scene graphs to construct multimodal knowledge graphs that support cross-modal reasoning [155].

The integration of these structural representations enables what researchers term structure-aware reasoning, where the system respects the original document layout to answer complex queries. DMAP explicitly encodes hierarchical organization and inter-element relationships, creating a human-aligned schema that enhances retrieval precision and reasoning consistency compared to conventional approaches [2]. Similarly, SuperRAG utilizes layout-aware graph modeling to retain the connections between text chunks, tables, and figures, allowing for the handling of complex questions that require synthesizing information across modalities [55]. For long documents, frameworks like HiKEY and LAD-RAG reconstruct logical heterogeneous graphs or symbolic document graphs to capture cross-page dependencies, ensuring that evidence retrieval is not limited to isolated snippets but reflects the document's global structure [31]. LAD-RAG specifically constructs a symbolic document graph during ingestion to capture layout structure and cross-page dependencies, adding it alongside standard neural embeddings to yield a more holistic representation [32].

Despite the progress, contradictions and challenges remain regarding the optimal balance between visual and structural representations. While some studies argue that document-as-image representations are suboptimal for text-rich scientific retrieval compared to interleaved text-image formats derived from source code like LaTeX, others contend that image-based approaches are superior for preserving the holistic context of visually rich enterprise documents [172].

Additionally, the aggregation of hundreds of visual patch tokens into single vectors for efficiency can lead to the collapse of distinct semantic details, particularly in financial documents where minor visual differences carry significant meaning [111]. Consequently, the field is moving toward adaptive systems like AutoThinkRAG, which utilize complexity-aware routing to dynamically select between visual and structural reasoning paths, ensuring that the representation strategy aligns with the specific demands of the query [169]. This evolution signifies a broader transition from static, one-size-fits-all parsing to dynamic, context-sensitive frameworks that unify visual perception with structural logic.

Architectural Paradigms and Retrieval Mechanisms

The architectural landscape of Multimodal Retrieval-Augmented Generation (RAG) is undergoing a fundamental transformation, shifting from linear retrieve-then-generate pipelines toward sophisticated, agentic frameworks capable of dynamic reasoning and multi-step decision-making. A primary divergence in current literature exists between unified embedding strategies that treat documents holistically as images and modular approaches that decouple retrieval, reasoning, and routing to preserve structural fidelity. Early multimodal RAG systems often relied on converting visual content into text summaries before embedding, a process that identifies as a source of significant information loss regarding spatial relationships and numerical precision [53]. In contrast, direct multimodal embedding retrieval, where images are stored natively in the vector space, has been shown to outperform text-based summarization approaches by substantial margins in both retrieval precision and answer accuracy. This shift is further supported by findings that bypassing traditional parsing to embed documents directly as images via Vision-Language Models can yield end-to-end performance gains of 20-40% over text-based pipelines by maximizing the retention of layout and visual cues [113].

Despite the efficacy of unified embeddings, a significant body of research argues for modular architectures that explicitly model document structure and cross-modal dependencies. A comprehensive survey categorizes these modular roles into Modality-Unifying Captioners, Multimodal Embedders, and End-to-End Representers, highlighting trade-offs in latency, index size, and information fidelity [7]. To address the fragmentation caused by flat chunking, several frameworks introduce graph-based representations to link visual entities with textual concepts. Proposals include constructing multimodal knowledge graphs that enable reasoning along explicit logical chains rather than relying solely on semantic similarity [155]. Similarly, other approaches construct a symbolic document graph alongside neural embeddings to capture layout structures and cross-page dependencies, allowing an LLM agent to dynamically retrieve evidence based on query demands rather than static top-k limits [32]. This structural awareness is critical for visually rich documents, where benchmarks reveal that multimodal text-image fusion systems consistently outperform both unimodal and jointly multimodal embedding-based retrieval, suggesting that current unified embeddings remain inadequate for complex document-centric tasks [26].

The evolution toward agentic frameworks represents the next frontier, where systems employ reflective reasoning and iterative workflows to handle complex, multi-hop queries. Recent introductions include multi-agent frameworks utilizing an iterative workflow of exploration, summarization, and reflection to enhance reasoning capabilities in dense visual documents [80]. This aligns with systems that employ a consistency-constrained modality fusion mechanism to align reasoning processes across textual and visual pipelines at inference time, significantly improving accuracy when critical information is distributed across modalities [9]. Furthermore, some architectures propose a perception-reasoning decoupling where a lightweight VLM acts as a visual interpreter, passing textual representations to an LLM for logical synthesis, thereby optimizing efficiency and robustness for complex queries [169]. These agentic systems often incorporate self-correction mechanisms, such as agents that dynamically assess the sufficiency of retrieved context against a structural document map, iteratively refining answers to ensure evidence-driven consistency [2].

Contradictions and nuances persist regarding the optimal granularity of retrieval and the balance between visual and textual processing. While some frameworks advocate for page-level retrieval enhanced by dual-cue filtering to reduce computational overhead [49], others argue that page-level retrieval is too coarse, proposing a block-level approach that segments documents based on layout to achieve higher precision and reduce token consumption by over 70% [8]. Additionally, evaluations highlight a persistent text-over-visual modality bias in current MLLMs, noting that even with ground-truth retrieval, models struggle with chart-based reasoning, indicating that unified embedding methods often fail in specialized visual scenarios [3]. The debate extends to evaluation, where studies emphasize that existing benchmarks often overlook fine-grained visual grounding and multi-document synthesis, leading to an over-confidence dilemma where systems generate answers without sufficient evidence support [95]. Consequently, the field is moving toward hybrid architectures that reconceptualize multimodal content as interconnected knowledge entities using dual-graph construction to navigate both structural and semantic relationships, aiming to eliminate the architectural fragmentation that constrains current systems [46].

Hierarchical, Graph-Based, and Layout-Oriented Indexing

To effectively manage the complexity of long documents and preserve their structural integrity, contemporary research has shifted away from flat text chunking toward sophisticated hierarchical tree-based and graph-based indexing strategies. This paradigm shift addresses the fundamental limitation of traditional Retrieval-Augmented Generation (RAG) systems, which often fragment documents into isolated passages, thereby severing the logical and spatial dependencies essential for accurate comprehension. Frameworks such as HiKEY and LAD-RAG exemplify this transition by reconstructing logical heterogeneous graphs or symbolic document graphs that explicitly encode parent-child relationships. These structures enable coarse-to-fine retrieval strategies that prune the search space efficiently before engaging in fine-grained extraction. For instance, HiKEY utilizes Document Hierarchical Parsing to elevate document hierarchy to a first-class retrieval signal, performing global routing to rapidly eliminate irrelevant sections before ranking evidence through multimodal fusion [31]. Similarly, LAD-RAG constructs a symbolic document graph during ingestion to capture layout structures and cross-page dependencies, allowing an LLM agent to dynamically interact with both neural and symbolic indices for adaptive evidence retrieval [32].

The necessity for such structural preservation is particularly acute in semi-structured and visually rich documents where elements like tables, charts, and hierarchical paragraphs are interleaved in irregular layouts. MoDora addresses the fragmentation of OCR-parsed elements by adopting a local-alignment aggregation strategy and designing a Component-Correlation Tree (CCTree) to hierarchically organize components, explicitly modeling inter-component relations through a bottom-up cascade summarization process [294]. This approach contrasts with methods that rely solely on token-level semantics, which often fail in reasoning across document elements such as quantities and dates. Doc2SoarGraph tackles this by employing semantic-oriented hierarchical graph structures to capture correlations among different elements within table-text documents, significantly outperforming conventional methods in discrete reasoning tasks [151]. Furthermore, the integration of layout information is critical; layout-oriented approaches perform segmentation to construct fine-grained retrieval units, integrating local semantics with global context to reduce token consumption and improve answer accuracy. LFRAG advances this by moving from page-level to block-level retrieval, utilizing a semantic-layout fusion encoder to align queries precisely with content blocks, thereby reducing irrelevant context in downstream generation [8].

Beyond explicit tree and graph constructions, other methodologies leverage inherent document topologies or generate pseudo-structures to enhance retrieval. LARAG demonstrates that leveraging author-defined hyperlink structures in technical documentation can enable an implicit form of graph-like retrieval without explicit graph construction, yielding more faithful and efficient pipelines [275]. In scenarios where explicit structure is absent or difficult to parse, systems like PG-RAG conceptualize LLMs as students that autonomously record factual information to form interconnected "mental indices" or pseudo-graph databases, mimicking human note-taking behaviors to refine retrieval paths [166]. Similarly, DocsRay generates a hierarchical pseudo-Table of Contents to guide zero-shot retrieval, effectively reducing retrieval complexity while maintaining high accuracy across diverse document elements [25]. The efficacy of these hierarchical approaches is further supported by MultiDocFusion, which integrates vision-based parsing with LLM-based section parsing to reconstruct document structures into hierarchical trees, demonstrating that structure-aware chunking significantly improves retrieval precision and answer quality in industrial settings [52].

Contradictions and agreements exist regarding the optimal granularity and modality for these indexing strategies. While some approaches like ColPali and VisionRAG advocate for vision-first retrieval that indexes document images directly to preserve spatial cues without brittle OCR pipelines, others argue for the necessity of explicit structural parsing to handle complex reasoning [233; 114]. However, there is a strong consensus that flat chunking is insufficient for multimodal long-context documents. \$G^2\$-Reader introduces a dual-graph system comprising a Content Graph for document-native structure and a Planning Graph for tracking intermediate findings, addressing the brittleness of iterative retrieval in long contexts where noise accumulation often leads to drift [2]. This aligns with findings from DMAP, which posits that flattening documents disrupts logical dependencies that humans naturally exploit, and thus proposes a human-aligned hierarchical schema to capture both semantic and layout dependencies [2]. Additionally, the use of knowledge graphs extends beyond single documents; RAKG and Docs2KG focus on constructing document-level or unified knowledge graphs from

heterogeneous sources to capture global information and interconnections, mitigating the issue of long-context forgetting in LLMs [280; 196].

The evolution of these indexing mechanisms also intersects with generative retrieval and advanced parsing techniques. GenRet learns to tokenize documents into short discrete representations via discrete auto-encoding, offering an end-to-end alternative to rule-based identifiers that often fail to capture complete semantic information [249]. Meanwhile, comprehensive surveys on document parsing highlight the transition from modular pipeline systems to end-to-end vision-language models, emphasizing that reliable structured data extraction is foundational for next-generation intelligent systems [224]. The integration of these parsing capabilities with hierarchical indexing is evident in systems like ZenDB, which leverages semantic hierarchical structures extracted from templated documents to answer ad-hoc SQL queries, achieving significant cost savings and accuracy improvements over standard RAG baselines [286]. Ultimately, the convergence of hierarchical, graph-based, and layout-oriented indexing represents a critical advancement in document AI, enabling systems to navigate the intricate topologies of real-world documents with greater fidelity and efficiency.

Agentic, Dynamic, and Adaptive Retrieval Workflows

Static retrieval methods, which rely on query-agnostic document representations and fixed top- k selection, frequently fail to capture salient content in complex, multi-page documents, leading to incomplete evidence and degraded reasoning performance. This limitation has catalyzed the development of agentic, dynamic, and adaptive retrieval workflows that treat retrieval not as a one-time preprocessing step, but as an iterative, query-dependent process integrated deeply with reasoning. Systems like LAD-RAG address the structural deficiencies of conventional RAG in visually rich documents by constructing a symbolic document graph alongside neural embeddings, allowing an LLM agent to dynamically interact with both indices to fetch evidence based on layout and cross-page dependencies [32]. Similarly, SimpleDoc employs a lightweight agent that iteratively invokes a dual-cue retriever, refining the working memory with fresh pages until the query is confidently answered, thereby reducing the number of retrieved pages while improving accuracy [49]. These approaches underscore a paradigm shift from passive retrieval to active evidence gathering, where the retrieval strategy evolves in response to intermediate reasoning steps.

The necessity for adaptivity extends beyond simple iteration to include sophisticated mechanisms for assessing evidence sufficiency and reweighting document representations. The MARA framework introduces a Query-Aligned Region Encoder that constructs multi-level document representations reweighted by query relevance, coupled with a Self-Reflective Evidence Controller that monitors generation to dynamically incorporate content from lower-ranked sources if initial evidence is deemed insufficient [22]. This self-reflective capability is echoed in SLEUTH, a multi-agent framework that orchestrates a coarse-to-fine process to filter salient visual evidence and synthesize a distilled, evidence-dense context, effectively mitigating the redundancy often found in standard retrieved results [101]. Furthermore, DR-RAG proposes a two-stage dynamic relevance framework where a compact classifier determines the contribution of retrieved documents, enabling the system to mine relevance by combining document parts with the query without requiring multiple LLM calls, thus balancing efficiency and recall [211].

Contradictions and agreements arise when comparing static embedding approaches with these emerging dynamic paradigms. While traditional late-interaction retrievers like ColPali generate document embeddings independent of the query, newer models such as Argus-Retriever introduce query-conditioned late-interaction mechanisms. Argus utilizes a region-aware Mixture-of-Experts module where the document representation becomes dependent on the query, significantly improving retrieval scores on visual document benchmarks compared to query-agnostic baselines [65]. This aligns with the findings in RegionRAG, which argues that retrieving at the document level introduces substantial irrelevant visual content; instead, RegionRAG shifts the paradigm to region-level retrieval, dynamically grouping salient patches into semantic regions to focus the generator on concise, query-relevant content [109]. The agreement across these studies is clear: static, coarse-grained representations are insufficient for complex multimodal reasoning, and fine-grained, query-aware adaptation is essential.

Agentic workflows also demonstrate superior performance in handling logical dependencies and multi-hop reasoning that static pipelines often miss. MoLoRAG constructs a page graph to capture contextual relationships, enabling a lightweight VLM to traverse logical connections that semantic similarity alone would overlook [44]. In a similar vein, MDocAgent leverages five specialized agents to collaboratively retrieve and synthesize textual and visual cues, achieving significant improvements over methods that prioritize a single modality [279]. The effectiveness of iterative refinement is further validated by ViDoRAG, which employs an iterative agent workflow incorporating exploration, summarization, and reflection to handle complex reasoning across visual documents, outperforming existing methods by over 10% on competitive benchmarks [80]. Additionally, frameworks like DocLens utilize tool-augmented agents to "zoom in" on specific visual elements within retrieved pages, solving the dual-level failure of evidence localization at both page and element scales [75].

Despite the promise of agentic systems, challenges regarding efficiency and evaluation persist. While iterative agents improve accuracy, they can incur higher computational costs, prompting solutions like AVIR, which uses adaptive clustering and relevance thresholds to reduce the average page count required for answering by 70% without sacrificing performance [132]. Moreover, the evaluation of these dynamic systems remains a critical bottleneck, as noted in critiques of current benchmarks that often rely on synthetic data or incomplete ground truth, failing to capture the nuances of multi-source synthesis and agentic decision-making [95]. The emergence of frameworks like FAILURE IS FEEDBACK highlights the need for history-aware backtracking mechanisms that allow agents to learn from failed

traversals and dynamically manage the trade-off between retrieval accuracy and inference costs [182]. Collectively, these works illustrate that the future of document understanding lies in adaptive, agentic architectures that can dynamically navigate, refine, and reason over heterogeneous document structures, moving decisively away from the limitations of static, query-agnostic retrieval.

The Debate: Vision-First vs. Text-First Retrieval

A central tension in modern document intelligence lies in determining the optimal modality for the initial retrieval stage: preserving the document as a visual artifact or converting it into a textual representation. Proponents of vision-first approaches argue that documents are inherently multimodal structures where meaning is derived not just from characters, but from spatial layout, typography, tables, and figures. Traditional text-first workflows, which rely on Optical Character Recognition (OCR) followed by heuristic chunking, often discard these visuo-spatial cues and introduce brittleness to layout shifts [233] [114]. In response, models like ColPali and VisRAG have emerged, embedding document pages directly as images. These systems utilize late-interaction mechanisms to align query tokens with visual patches, demonstrating superior performance on benchmarks like ViDoRe and MMDocIR, particularly for tasks requiring the localization of information within complex layouts [233] [113] [57]. Empirical evidence suggests that visual retrievers can significantly outperform their text-based counterparts when the retrieval task depends on understanding the interplay between text and graphical elements [57].

However, the supremacy of vision-first methods is not absolute, and compelling evidence suggests that high-quality text representation remains highly competitive, and in some domains, superior. Research indicates that the performance gap often attributed to the retrieval mechanism may actually stem from inadequate document representation; when transcription and preprocessing are optimized, lexical retrievers like BM25 can recover much of the lost performance on multilingual and visually rich benchmarks [39]. Furthermore, the suitability of image-based representations varies by domain. In scientific retrieval, where critical evidence is distributed across structured sources like LaTeX equations and tables, document-as-image representations have been found to be consistently suboptimal compared to text-based representations, especially as document length increases [172]. This suggests that for text-dense, structurally rigorous documents, accessing the underlying source text or high-fidelity OCR provides a more robust signal than visual embeddings. Additionally, concerns regarding generalization persist; while vision-based models excel on documents similar to their training distribution, OCR-based pipelines often demonstrate superior adaptability to unseen documents of varying quality and degradation levels [127].

The choice between these paradigms is further complicated by document structure and query type. For instance, layout-aware parsing is deemed crucial for vertically structured reports where hierarchical relationships define meaning, whereas holistic page representations may suffice for horizontally structured slide decks [35]. This nuance has driven the development of hybrid and adaptive strategies that seek to balance the robustness of lexical search with the structural awareness of vision models. Systems like VisionRAG propose OCR-free, summary-guided multimodal retrieval that constructs lightweight semantic vectors using global page summaries and visual hotspots, aiming to match the efficiency of patch-based approaches while remaining adaptable to different encoders [114]. Similarly, frameworks such as HEAVEN introduce a two-stage hybrid-vector approach, using single-vector methods for efficient candidate retrieval over visually summarized pages before reranking with multi-vector methods to ensure accuracy [154]. Other approaches, like AgenticOCR, shift the paradigm from static full-document parsing to dynamic, query-driven extraction, effectively decoupling retrieval granularity from rigid page-level chunking to reduce visual token consumption and hallucination risks [117].

Despite advancements, significant challenges remain, particularly regarding multilingual capabilities and fine-grained grounding. Benchmarks such as MIRACL-VISION reveal a substantial gap in state-of-the-art VLM-based embedding models regarding multilingual retrieval, with some models performing markedly worse than text-based models in non-English contexts [175]. Furthermore, while visual retrievers excel at capturing layout, they can struggle with fine-print knowledge or small text details that OCR might capture more reliably, leading to proposals for holistic frameworks that explicitly integrate both salient and fine-print knowledge streams [42]. The nature of the query also dictates the optimal approach; while visual models handle extractive and layout-dependent queries well, they may falter on complex cross-document reasoning tasks where textual coherence is paramount [137]. Consequently, the field is moving towards a synthesis where the distinction between vision-first and text-first is blurred by architectures that leverage the strengths of both, such as using VLMs to generate rich textual descriptions for zero-shot retrieval or employing layout-informed multi-vector representations that fuse global page vectors with local sub-image embeddings [119] [271].

Specialized Modalities: Charts, Tables, and Diagrams

Charts and diagrams present unique challenges due to their dense semantic content and reliance on visual patterns for interpretation, necessitating a departure from standard retrieval methods that often fail to capture embedded statistical properties and practical applications. The fundamental limitation of existing text-to-chart retrieval solutions lies in the scarcity of comprehensive metadata, which hinders the understanding of both visual patterns and contextual information. To address this, recent research has focused on synthesizing hierarchical semantic insights that cover visual-oriented patterns, statistics-oriented properties, and task-oriented applications. For instance, a pipeline that automatically generates over 207,000 semantic insights to train CLIP-based models significantly outperforms state-of-the-art methods in both precise and fuzzy query scenarios [28]. This approach underscores the necessity of bridging the gap between visual patterns and textual queries through structured representation, moving beyond simple keyword matching to a deeper semantic alignment.

Complementing semantic synthesis, structured representation frameworks have emerged to model the intricate relationships within chart images. A knowledge graph-based approach unifies visual elements, encodings, and semantic relations, moving beyond simple raw data extraction to preserve the visual logic of the chart [98]. Similarly, reformulating chart data into Structured Triplet Representations reduces the gap between chart perception and reasoning tasks [253]. These structured approaches are critical for enabling systems to perform not just data lookup but also exploratory analysis. Furthermore, designing a joint multimodal scene graph explicitly represents relationships between chart elements, enhancing the understanding of structural and semantical knowledge which is often lost in pixel-based approaches [107]. By treating charts as structured graphs rather than mere images, these methods facilitate more robust reasoning capabilities.

The complexity of chart understanding is further exacerbated in real-world scientific and professional documents, where charts often require complex reasoning and common-sense understanding that synthetic datasets fail to capture. A benchmark based on real-world scientific literature highlights the limitations of synthetic data and evaluates first-order logic capabilities [239]. This aligns with findings that most existing datasets treat chart question answering in isolation, ignoring the broader document context and common-sense reasoning required for authentic queries [190]. Consequently, there is a growing consensus that effective systems must operate at the document level, integrating layout analysis with chart understanding, as demonstrated by a unified system for multimodal document parsing that preserves visual semantics across multi-granular representations [82].

Despite advancements in large multimodal models, significant gaps remain in graphical perception and reasoning. Research reveals that even state-of-the-art models struggle with basic perception tasks, such as generalizing across chart types and cross-referencing values, indicating that scaling model size alone is insufficient [259]. This is corroborated by benchmarks finding that performance varies significantly across different visual encoding channels and does not necessarily improve with model scale [235]. To mitigate these issues, specialized training and tool-integrated reasoning are becoming essential. A framework grounded in Tool-Integrated Reasoning decomposes complex analysis into observable steps using tools like OCR and instance segmentation, providing traceable evidence packages that improve robustness in sparse-annotation settings [277]. Similarly, distilling essential information from complex charts improves reasoning accuracy without requiring additional annotations [201].

The generation and summarization of charts also benefit from specialized modalities that link text and visual data. New datasets and models are capable of generating captions that describe perceptual and cognitive phenomena, going beyond simple data reporting [214]. In the realm of interactive documents, methods to create meaningful connections between textual phrases and visual channels facilitate reading comprehension [104]. Moreover, utilizing chain-of-thought strategies improves the logical coherence of chart summaries, addressing deficiencies in visual-language matching [67]. These developments suggest that future systems must not only retrieve and interpret charts but also synthesize them dynamically to support decision-making, as seen in frameworks that generate multi-subchart infographics from text-heavy documents by defining detailed metadata [185].

Finally, the evaluation of these specialized modalities requires benchmarks that reflect real-world complexity and diversity. Comprehensive evaluation sets covering diverse chart types and disciplinary topics reveal that current models still lag behind human performance in complicated reasoning tasks [282]. Additionally, new benchmarks address the

neglect of multi-hop reasoning required to integrate information from multiple charts, a common scenario in practical applications [58]. The push for more realistic evaluation is also evident in perspectives from scientific literature, which emphasize the need to include flowcharts and complex multi-plot figures to avoid inflated performance scores [236]. Collectively, these works illustrate a shift towards specialized, structured, and context-aware modalities that can handle the dense semantic content and visual intricacies of modern data visualization.

Semantic Synthesis and Knowledge Graph Representations

The evolution of chart understanding has shifted decisively from treating visualizations as mere containers for underlying tabular data toward rich, semantically synthesized representations that capture the intricate interplay between visual encodings and meaning. Early approaches often suffered from significant information loss by focusing exclusively on optical character recognition and raw data extraction, ignoring the visual semantics that convey the chart's true message. To bridge this gap, recent scholarship advocates for knowledge graph-based representations that unify visual elements with their semantic contexts. A pioneering effort in this domain is ChartKG, which proposes a novel framework to convert chart images into knowledge graphs by modeling visual elements, encodings, and insights in a unified manner. By integrating computer vision techniques with rule-based graph construction, ChartKG facilitates downstream applications like semantic-aware retrieval and question answering, effectively connecting pixel-level data to high-level semantic understanding. This approach addresses the limitation of existing methods that primarily focus on converting chart images into raw data, often neglecting visual encodings and semantic meanings essential for tasks like knowledge graph completion.

Parallel to structural graph modeling, there is a growing emphasis on enriching chart metadata through synthesized semantic insights to improve retrieval performance, particularly for the fuzzy queries common in Business Intelligence. The work Boosting Text-to-Chart Retrieval through Training with Synthesized Semantic Insights introduces a pipeline that automatically generates hierarchical insights covering visual patterns, statistical properties, and task-oriented applications, producing over 200,000 semantic insights for tens of thousands of charts. This approach underpins the training of models that leverage these synthesized insights to learn robust chart representations, demonstrating significant improvements over state-of-the-art methods in both precise and exploratory query scenarios. Similarly, ChartCards extends this concept by proposing a unified metadata generation framework that synthesizes data tables, visualization code, and multidimensional semantic captions. This framework supports the creation of large-scale datasets that enable multi-task fine-tuning, yielding substantial performance gains in text-to-chart retrieval and chart-to-table conversion tasks, highlighting the necessity of comprehensive metadata that goes beyond simple labels.

The integration of graph structures extends beyond static metadata to enable dynamic reasoning and deeper scene understanding. MSG-Chart designs a joint multimodal scene graph comprising visual and textual components to explicitly capture relationships between chart elements and underlying data patterns. This graph module serves as an inductive bias for vision transformers, enhancing performance on benchmark tasks by improving the model's grasp of structural and semantical knowledge. In the realm of unifying perception and reasoning, StructChart reformulates chart data into Structured Triplet Representations to reduce the gap between extracting visual information and performing logical reasoning. This approach is supported by novel metrics for evaluating chart-oriented representations, pushing the frontier of visual chart understanding beyond simple question answering. Furthermore, the application of knowledge graphs is not limited to individual charts but scales to heterogeneous document collections. Docs2KG presents a framework for extracting multimodal information from diverse unstructured documents to dynamically generate unified knowledge graphs, enabling efficient querying and exploration of enterprise data lakes.

Despite these advancements, challenges remain in aligning textual narratives with visual data and ensuring the fidelity of generated insights. ChartText addresses the difficulty of linking text with visualizations in interactive documents by employing a two-stage matching and grouping process, which facilitates reading comprehension and real-time annotation. However, the reliability of automated systems is often compromised by hallucinations or shallow analysis. EvidFuse proposes a training-free multi-agent framework that enables writing-time interleaved generation of text and charts, ensuring that visual evidence is constructed on-demand to support narrative claims, thereby mitigating insight freezing and inconsistency. Additionally, the complexity of scientific charts often demands deeper reasoning capabilities. RealCQA introduces a benchmark based on real-world scientific literature to test first-order logic capabilities, revealing that existing models struggle with the visual complexity and structured output requirements of authentic scientific charts.

The synthesis of semantic knowledge also plays a pivotal role in Retrieval-Augmented Generation systems tailored for visually rich documents. Traditional pipelines often fail to capture cross-modal dependencies, leading to incomplete retrieval. To overcome this, MMGraphRAG refines visual content through scene graphs to construct multimodal knowledge graphs, employing spectral clustering for cross-modal entity linking and retrieving context along reasoning

paths. Similarly, MegaRAG incorporates visual cues into the construction and retrieval phases of knowledge graphs to enable cross-modal reasoning for long-form content, outperforming text-only solutions. The importance of structured representation is further echoed in VisText, which provides a dataset of semantically rich captions and scene graphs to improve the generation of descriptions that articulate perceptual and cognitive features of charts. Moreover, specialized approaches like SIMPLOT focus on distilling essential elements from complex charts to enhance reasoning accuracy, demonstrating that filtering irrelevant information is as crucial as enriching semantic content.

The trajectory of research indicates a convergence where semantic synthesis and graph-based representations are becoming foundational for advanced chart understanding. By moving beyond isolated visual features to model the intricate web of relations between data, visuals, and text, these methods enable more robust, interpretable, and context-aware systems. Whether through the explicit construction of knowledge graphs as seen in ChartKG and MMGraphRAG, or the synthesis of hierarchical insights for training as in Boosting Text-to-Chart Retrieval through Training with Synthesized Semantic Insights, the field is increasingly recognizing that true chart comprehension requires a holistic representation of both form and meaning. This shift is essential for supporting complex downstream tasks ranging from open-ended question answering to the generation of complex statistical infographics, ultimately fostering a new generation of tools capable of deep analytical reasoning over visual data.

Scientific Diagram Analysis and Generation

The capacity to interpret and generate scientific diagrams represents a pivotal frontier in multimodal document understanding, distinguishing itself from standard chart analysis by requiring the comprehension of abstract logical workflows and schematic relationships rather than mere quantitative data extraction. While Large Language Models (LLMs) and Multimodal LLMs (MLLMs) have achieved proficiency in processing text and basic plots, they frequently falter when confronted with the nuanced conceptual structures found in scientific teaser figures and flowcharts. This limitation is largely attributed to a scarcity of training data that aligns complex diagrams with their textual context and authorial intent. To bridge this gap, recent initiatives have focused on curating large-scale datasets that pair schematic diagrams with rich metadata, enabling retrieval-augmented generation (RAG) systems to perform exemplar-driven synthesis. For instance, the introduction of DiagramBank provides a vast collection of schematic diagrams curated from top-tier publications, paired with abstracts and in-text references, thereby facilitating the generation of publication-grade teaser figures that serve as strategic visual interfaces for scientific narratives [45]. Similarly, the mPLUG-PaperOwl framework addresses the weakness of MLLMs in diagram analysis by constructing the M-Paper dataset, which aligns diagrams with related paragraphs and introduces user "outlines" as control signals to better synchronize model outputs with the specific logical flow intended by authors [38].

Beyond generation, the accurate analysis of scientific diagrams demands benchmarks that reflect the inherent complexity of real-world literature, moving away from synthetic or template-based evaluations that often yield inflated performance scores. Traditional benchmarks have been criticized for relying on simplistic visual elements and narrow chart types, failing to capture the intricate multi-plot figures and structural diagrams prevalent in scientific discourse [236]. In response, new evaluation frameworks like SCI-CQA and RealCQA have been developed to test models on flowcharts and real-world charts extracted from scientific papers, emphasizing first-order logic and complex reasoning capabilities [239]. These efforts reveal that current models often struggle with the visual complexity and contextual dependencies of authentic scientific figures, particularly when tasks require joint comprehension of multiple diagrams or the integration of visual and textual evidence [99]. Furthermore, the difficulty of statistical reasoning over charts is highlighted by the CLIMATEVIZ benchmark, where even state-of-the-art models significantly underperform compared to human experts in fact-checking claims grounded in scientific visualizations [178].

To overcome these analytical hurdles, researchers are exploring diverse architectural paradigms that move beyond black-box predictions toward transparent, reproducible workflows. One promising direction involves tool-integrated reasoning, where agents dynamically orchestrate specialized tools for element detection, segmentation, and optical character recognition to systematically parse complex charts, as demonstrated by the ChartAgent framework [277]. Another approach reformulates data extraction tasks, such as parsing line charts, as instance segmentation problems to robustly handle occlusions and structural variations that confound traditional keypoint-based methods [281]. Additionally, the use of knowledge graphs offers a structured representation of visual elements and semantic relations, facilitating downstream tasks like retrieval and question answering by modeling the intricate connections within chart images [98]. Code-guided synthesis is also gaining traction, with pipelines like ChartGen leveraging code reconstruction to generate synthetic chart-image pairs, thereby scaling up training data and improving model robustness across diverse chart types [285].

The interplay between text and diagrams remains a critical area of investigation, as scientific arguments are constructed through the tight coupling of narrative and visual evidence. Systems like EvidFuse and ChartText aim to resolve inconsistencies between text and charts by enabling writing-time evidence learning and automated linking of text phrases to visual elements, ensuring that narratives evolve alongside the visual evidence they describe [183; 104]. However, achieving robust alignment is challenging, especially when models must verify claims grounded in statistical reasoning or navigate long documents where evidence is dispersed. Benchmarks like BRIDGE and M4DocBench expose systematic deficiencies in current multimodal systems regarding evidence aggregation and grounding across long contexts, underscoring the need for models that can perform multi-hop reasoning across text, tables, and figures [217; 82]. Furthermore, the generation of complex infographics from text-heavy documents requires deep content understanding, a capability being advanced by frameworks like Infogen, which utilize two-stage processes to generate metadata and convert it into contextually accurate infographic code [185]. Despite these advances, significant gaps remain in handling multi-modal reasoning across diverse domains and ensuring that generated visuals are not only

aesthetically pleasing but also logically consistent with the underlying data and narrative, a challenge that continues to drive the development of specialized datasets and evaluation frameworks.

Integration of Charts into General Document RAG

Integrating chart understanding into general document Retrieval-Augmented Generation (RAG) pipelines necessitates a fundamental departure from flat text chunking toward layout-aware, multimodal processing strategies that preserve the structural integrity of visually rich documents. Traditional RAG systems often rely on Optical Character Recognition (OCR) to convert visual elements into text, a process that frequently results in the loss of spatial relationships, numerical precision, and the contextual links between narrative text and graphical data. To mitigate this information loss, advanced frameworks now explicitly group table and figure images for batch processing by multimodal Large Language Models (MLLMs), generating structured summaries that are indexed alongside narrative text. For instance, specific financial frameworks employ a tiered fallback strategy where lightweight MLLMs produce structured JSON outputs and concise textual summaries from charts and tables, ensuring quantitative data remains accessible for cross-modal reasoning without overwhelming the context window [47]. This approach ensures that quantitative data within charts is not lost during the text extraction phase and remains accessible for complex reasoning tasks.

The challenge of handling mixed layouts where text, tables, and charts coexist is further addressed through graph-based modeling and hierarchical structuring, which explicitly retain the connections between disparate document elements. Layout-aware graph modeling has been introduced to capture the relationships between text chunks, tables, and figures, enabling systems to handle complex queries requiring information synthesis across multiple modalities [55]. This approach aligns with dynamic RAG frameworks that construct a symbolic document graph during ingestion to capture layout structure and cross-page dependencies, allowing an LLM agent to dynamically retrieve evidence based on query demands rather than static top-k retrieval [32]. Furthermore, tree-based systems utilize a Component-Correlation Tree to hierarchically organize semi-structured document elements, explicitly modeling inter-component relations to link descriptive paragraphs with distant table cells or charts [294]. These structural representations are critical because standard embedding models often struggle to capture the logical chains between modalities, a gap that multimodal knowledge graph approaches attempt to fill by refining visual content through scene graphs and constructing interpretable knowledge structures [155].

Despite these architectural advances, the fidelity of chart understanding within RAG systems depends heavily on the quality of visual parsing and the ability to link textual narratives with visual data. Some frameworks address the degradation of MLLM performance in sparse-annotation settings by employing Tool-Integrated Reasoning, where an agent orchestrates tools like instance segmentation and OCR to create a traceable Evidence Package for chart analysis [277]. Complementing this, other methods focus on linking text with visualizations by matching phrases to chart elements and grouping them according to visual channels, facilitating reading comprehension in interactive documents [104]. The importance of such linking is underscored by findings in multi-document QA systems, which demonstrate that a consistency-constrained modality fusion mechanism significantly enhances accuracy when critical information is distributed across text and visuals [9]. However, contradictions exist regarding the optimal retrieval modality; while some studies advocate for direct multimodal embedding retrieval to preserve visual context, showing significant improvements over text-based summarization approaches [53], others suggest that unified multimodal text-image fusion RAG systems outperform jointly multimodal embedding-based retrieval, indicating that current embedding techniques may still be inadequate for complex chart scenarios [26].

Moreover, the integration of charts requires robust handling of diverse chart types and complex reasoning tasks that extend beyond simple data extraction. Benchmarks focusing on scientific literature highlight that existing models often fail on real-world charts containing flowcharts and multi-plot figures, necessitating evaluation frameworks that test first-order logic capabilities [236] [239]. To support such reasoning, frameworks synthesize comprehensive metadata including data tables and visual elements to enable multi-task understanding, while others generate multi-step visual reasoning data to train models on computation-intensive tasks [177] [116]. The necessity for deep contextual understanding is further emphasized by document-level chart question answering tasks, which frame the problem as requiring common-sense reasoning and layout analysis to locate charts before querying them [190]. Recent evaluations also reveal a consistent text-over-visual modality bias in current MLLMs during chart-based reasoning, suggesting that future integration strategies must actively force the model to attend to visual evidence rather than relying on textual priors [3]. Ultimately, the convergence of layout-aware parsing, graph-based structural modeling, and tool-integrated

reasoning represents the current frontier in ensuring that quantitative data within charts is not lost but effectively leveraged for holistic document intelligence.

Benchmarking Landscapes and Evaluation Metrics

The rapid evolution of Multimodal Retrieval-Augmented Generation (MRAG) has necessitated a paradigm shift in evaluation methodologies, moving from simple image-text pairing tasks to complex benchmarks that demand cross-modal synthesis, fine-grained evidence attribution, and robustness against distractors. Early benchmarks often treated visual and textual modalities in isolation or focused on single-document contexts, failing to capture the intricacies of real-world document understanding where information is distributed across tables, charts, and lengthy textual narratives. To address these limitations, recent literature has introduced comprehensive suites designed to evaluate multi-document reasoning and visually rich elements. For instance, VisDoM: Multi-Document QA with Visually Rich Elements Using Multimodal Retrieval-Augmented Generation introduces VisDoMBench, a benchmark specifically targeting multi-document settings with rich multimodal content, including presentation slides and charts, emphasizing the need for consistency-constrained modality fusion. Similarly, UniDoc-Bench: A Unified Benchmark for Document-Centric Multimodal RAG provides a large-scale evaluation framework derived from tens of thousands of real-world PDF pages, enabling apples-to-apples comparisons across text-only, image-only, and multimodal fusion paradigms, thereby highlighting the insufficiency of unimodal approaches in document-centric tasks.

A critical trend in current benchmarking is the specific focus on chart and table reasoning, which are prevalent in scientific and financial domains but often overlooked in generalist evaluations. Benchmarking Multimodal RAG through a Chart-based Document Question-Answering Generation Framework presents Chart-MRAG Bench, revealing that state-of-the-art models suffer from a text-over-visual bias and struggle with unified multimodal embedding retrieval in chart-based scenarios. This finding is corroborated by WikiMixQA: A Multimodal Benchmark for Question Answering over Tables and Charts, which demonstrates a significant performance drop in proprietary and open-source models when retrieval from long documents containing tables and charts is required, as opposed to direct context provision. Furthermore, ChartMind: A Comprehensive Benchmark for Complex Real-world Multimodal Chart Question Answering and ChartX & ChartVLM: A Versatile Benchmark and Foundation Model for Complicated Chart Reasoning argue that existing evaluations rely too heavily on rigid output formats, advocating for benchmarks that support open-domain textual outputs and diverse chart formats to better reflect real-world analytical demands. The complexity is further elevated in MultiChartQA: Benchmarking Vision-Language Models on Multi-Chart Problems, which introduces tasks requiring comparative and sequential reasoning across multiple charts, exposing the inability of current models to perform multi-hop reasoning over disjointed visual evidence. Specialized domains such as finance are also gaining traction with benchmarks like FinDocMRE: A Benchmark for Document-Level Financial Multimodal Reasoning Evaluation, which reveal that models struggle significantly with numerical estimation and cross-page visual grounding in professional contexts.

Beyond domain-specific challenges, there is a growing emphasis on diagnostic tools for assessing hallucination, faithfulness, and the precise attribution of evidence. The field is increasingly recognizing that answer correctness alone is an insufficient metric; the provenance of the generated information must be verifiable. MRAG-Suite: A Cross-Domain Diagnostic Benchmark and Tooling for Visual Retrieval-Augmented Generation addresses this by introducing a claim-level diagnostic judge that evaluates hallucination, faithfulness, and cross-modality agreement, revealing strong modality biases and the under-utilization of retrieved evidence in current systems. Complementing this, FATHOMS-RAG: A Framework for the Assessment of Thinking and Observation in Multimodal Systems that use Retrieval Augmented Generation proposes a phrase-level recall metric and a nearest-neighbor embedding classifier to identify pipeline hallucinations, finding that closed-source pipelines significantly outperform open-source ones in correctness and hallucination metrics. The necessity for fine-grained attribution is further explored in Look as You Think: Unifying Reasoning and Visual Evidence Attribution for Verifiable Document RAG via Reinforcement Learning, which introduces the Chain-of-Evidence paradigm to ground reasoning steps to specific bounding boxes, and VISA: Retrieval Augmented Generation with Visual Source Attribution, which highlights the challenge of linking generated content to exact visual regions rather than just document-level references.

The evaluation landscape also increasingly scrutinizes the reasoning capabilities of models over long-context and multi-hop scenarios, where structural understanding is paramount. BRIDGE: Benchmark for multi-hop Reasoning In long multimodal Documents with Grounded Evidence and DocHop-QA: Towards Multi-Hop Reasoning over Multimodal Document Collections both emphasize that conventional answer-only evaluations hide systematic

deficiencies in evidence aggregation and grounding, particularly when reasoning requires integrating information across text, tables, and figures in long scientific papers. G^2 -Reader: Dual Evolving Graphs for Multimodal Document QA argues that flat chunking breaks document-native structure, proposing dual-graph systems to preserve cross-modal semantics and track intermediate findings. Additionally, LAD-RAG: Layout-aware Dynamic RAG for Visually-Rich Document Understanding and LFRAG: Layout-oriented Fine-grained Retrieval-Augmented Generation on Multimodal Document Understanding critique coarse-grained page-level retrieval, demonstrating that layout-aware, block-level retrieval significantly improves accuracy and efficiency by preserving structural dependencies. Contradictions and gaps remain, however; while Retrieval Meets Reasoning: Even High-school Textbook Knowledge Benefits Multimodal Reasoning suggests that simple retrieval scaffolds can boost reasoning, ARE WE ON THE RIGHT WAY FOR ASSESSING DOCUMENT RETRIEVAL-AUGMENTED GENERATION? warns of an "over-confidence dilemma" where frameworks provide answers without evidence support, urging for more rigorous, component-wise assessment. Finally, specialized evaluations for ESG tasks and multilingual settings, such as Benchmarking Multimodal Understanding and Complex Reasoning for ESG Tasks and VisR-Bench: An Empirical Study on Visual Retrieval-Augmented Generation for Multilingual Long Document Understanding, underscore the need for domain-specific and linguistically diverse evaluation standards to ensure robustness in global deployments.

Domain-Specific and Multilingual Evaluation Suites

The landscape of benchmarking for Retrieval-Augmented Generation (RAG) has shifted decisively from general-purpose text evaluation toward highly specialized suites designed to capture the unique structural and semantic complexities of professional domains. In the financial sector, where precision is paramount and data is often embedded in intricate layouts, recent benchmarks have exposed the limitations of text-only approaches. [14] addresses the critical oversight of visual content in financial analysis by introducing a suite that integrates multimodal data and enforces visual citation, ensuring that generated insights are traceable to specific charts or tables. This focus on traceability is complemented by [187], which presents FinTMMBench, a benchmark uniquely incorporating temporal awareness. This dataset requires models to interpret data across specific time periods using hybrid corpora of tables, news, and stock charts, reflecting the dynamic nature of financial decision-making. The difficulty of these tasks is further underscored by [10], which demonstrates that even advanced models struggle with nested tables and multi-dimensional plots without specialized pre-processing pipelines. Furthermore, the brittleness of current systems in non-English contexts is highlighted by [89], which reveals that while models perform adequately on text extraction, they exhibit significant failures in chart interpretation and multi-turn dialogue within French financial documents. The risk of information loss in these high-stakes environments is quantified in [111], which argues that single-vector aggregation strategies often collapse distinct financial documents into identical representations, obscuring critical semantic shifts caused by minor numerical changes. Additionally, [248] reveals that no current model surpasses an overall score of 65 on complex document-level reasoning, highlighting persistent challenges in integrating visual grounding with logical reasoning.

Beyond finance, the engineering and scientific domains face distinct challenges related to schematic diagrams, markup cues, and cross-document reasoning. [212] addresses a specific blind spot in scientific literature: the ability of models to treat visual markups like highlights and underlines as logical directives for table reasoning. In the realm of scientific claim verification, [229] provides a rigorous benchmark for verifying claims across diverse modalities within scientific papers, exposing substantial performance gaps between current models and human experts. The complexity of multi-document reasoning is further explored in [222], which mirrors real research workflows by requiring models to synthesize information across primary papers and their cited documents. Additionally, [72] pushes the boundaries of reasoning by constructing queries that require navigating information distributed across multiple scientific documents and modalities without explicit hyperlinks. These specialized suites collectively illustrate that robust performance requires evaluation frameworks that are as complex and diverse as the documents they aim to understand.

A critical and growing dimension of these evaluation suites is the emphasis on multilingual capabilities, which reveals significant disparities in model performance across languages. [175] covers 18 languages and demonstrates that state-of-the-art visual models suffer from severe performance degradation in non-English contexts, with retrieval accuracy dropping significantly compared to text-based models. This multilingual gap is corroborated by [37], which spans sixteen languages and highlights struggles with low-resource languages and structured tables. Specific efforts have been made to address underrepresented languages and scripts, such as [34], which focuses on the structural complexity of Korean official publications, and [144], which exposes the limitations of current models in handling Arabic script, calligraphy, and table structures. [137] further integrates multilingual splits across French, English, Spanish, and German to provide a more accurate assessment of non-English capabilities, noting that models trained exclusively on English data fail to generalize effectively.

The construction of these benchmarks increasingly relies on human-in-the-loop processes to ensure high-quality, realistic query distributions that avoid the pitfalls of purely synthetic data. [193] utilizes expert-annotated Q&A pairs across diverse domains to evaluate design choices in real-world scenarios, emphasizing the importance of data parsing. [48] expands on this by offering human-verified queries across ten datasets and six languages, focusing on fine-grained visual grounding and multi-document synthesis. Similarly, [12] employs a human-AI collaborative pipeline to validate question-answer pairs in the Environmental, Social, and Governance domain, ensuring that the benchmark captures the structural diversity of ESG reports. The necessity of such rigorous annotation is underscored by [95], which introduces DOUBLE-BENCH to provide fine-grained assessments of RAG components, revealing that current frameworks often suffer from over-confidence and lack of evidence support. These specialized suites collectively illustrate that while general capabilities have improved, robust performance in domain-specific and multilingual settings remains a significant frontier.

Conclusion

The literature reveals a decisive paradigm shift in document intelligence, moving from brittle, text-centric pipelines toward holistic, multimodal frameworks that preserve the structural and visual integrity of Visually Rich Documents. The consensus across recent studies indicates that traditional Optical Character Recognition-based workflows fundamentally disrupt the logical and spatial dependencies essential for accurate reasoning, particularly when dealing with complex tables, charts, and interleaved layouts. In response, the field is converging on unified visual representations and parsing-free methodologies that embed documents directly as images or structured graphs. This evolution addresses the critical issue of structural fragmentation, ensuring that layout cues, font styles, and graphical relationships are retained rather than discarded during the indexing phase. By treating documents as cohesive visual entities or explicit knowledge graphs, modern systems achieve superior fidelity in capturing the nuanced interplay between textual narratives and visual data.

Architecturally, the domain is transitioning from static, one-shot retrieval mechanisms to dynamic, agentic workflows capable of iterative reasoning and adaptive evidence gathering. The limitations of flat chunking have spurred the development of hierarchical, graph-based, and layout-aware indexing strategies that model cross-page dependencies and inter-element relationships. These structural advancements enable coarse-to-fine retrieval processes that efficiently prune search spaces while maintaining global context. Furthermore, the emergence of multi-agent frameworks introduces a layer of reflective reasoning, allowing systems to decompose complex queries, validate evidence sufficiency, and dynamically adjust search strategies. This move toward active, query-dependent retrieval marks a significant departure from passive information extraction, facilitating deeper, multi-hop reasoning capabilities that were previously unattainable with linear pipelines.

Despite these advancements, significant challenges persist regarding modality bias, evaluation rigor, and scalability. A recurring contradiction in the literature highlights a persistent text-over-visual bias, where models often under-utilize retrieved visual evidence in favor of textual heuristics, even when visual data is paramount for the correct answer. Additionally, the debate between vision-first and text-first approaches remains unresolved, with trade-offs emerging between the structural fidelity of image-based embeddings and the robustness of lexical search across diverse document qualities and languages. Current evaluation benchmarks frequently fail to capture the complexity of real-world scenarios, often relying on synthetic data that masks systemic deficiencies in grounding and faithfulness. Consequently, there is an urgent need for comprehensive, human-verified benchmarks that assess fine-grained visual attribution, cross-modal consistency, and performance in low-resource linguistic contexts.

Looking forward, the trajectory of Multimodal Retrieval-Augmented Generation points toward the integration of adaptive, hybrid architectures that seamlessly blend visual perception with structural logic. Future progress hinges on developing efficient indexing strategies that mitigate the computational overhead of high-dimensional visual embeddings without sacrificing granularity, as well as creating robust mechanisms for fine-grained visual citation and explainability. The ultimate goal is to establish systems that not only retrieve and generate accurate responses but also provide transparent, verifiable evidence chains grounded in specific visual regions. By addressing the bottlenecks of scalability, multilingual generalization, and modality alignment, the next generation of document intelligence systems will be better equipped to handle the heterogeneous and complex nature of real-world information, transforming how organizations interact with their most valuable data assets.